REALIZATION OF ARBITRARY MULTIPLE Q-ZETA STAR VALUES AS SIMPLE GRAPH SERIES

ANTUN MILAS

Allowed standard prerequisites (editorial).

- Standard q-binomial summation, vertex-algebra character definitions and classical elliptic/theta functions. Original source bibliography, including Lang, Arakawa and AM where relevant.

Source reference. Original reference 1.1: algebra-geometry/sources/sigma-2026-087/native/sigma26-087.tex, source lines 202–202. Cross-reference gives an introductory specialization, earlier narrative comparison or section pointer; all source-local arguments used for the counted theorem are included.

Our approach is motivated on the one hand by our previous work on arc spaces and graph series [15, 30, 31, 36] (see also [35]), and on the other hand by character formulas for certain vertex algebras introduced by Arakawa [5], related to genus-zero class-$\mathcal{S}$ theories in physics [10, 11]. As already pointed out in [15, 31, 35, 36], the Hilbert series of the arc algebra $J_\infty(R_\Gamma)$, where R_Γ is the edge algebra (defined below) of a simple[1] graph Γ, coincides with the character of a certain *principal subspace* W_Γ (after Feigin and Stoyanovsky [24]) inside an appropriate lattice vertex algebra [17, 32, 40]. Moreover, the arc algebra $J_\infty(R_\Gamma)$ carries the structure of a commutative vertex algebra, and this correspondence can be realized as an isomorphism of commutative graded vertex algebras. We stress, however, that the principal subspace W_Γ is not a conformal vertex algebra, so its grading is not canonical. Therefore, following [35], we equip the principal subspace W_Γ with the grading

$$\deg\big(\mathrm{e}^{\alpha_i}_{(-m-1)}\big) = m + b_i,$$

where

$$Y(\mathrm{e}^{\alpha_i}, x) = \sum_{m\in\mathbb{Z}} \mathrm{e}^{\alpha_i}_{(m)} x^{-m-1},$$

together with the color grading recording the multiplicity of each generator e^{α_i}. This grading is the analogue of the grading on the arc algebra $J_\infty(R_\Gamma)$, where the jet variables satisfy

$$\deg(x_{i,m}) = m + b_i.$$

Thus the correspondence

$$x_{i,m} \longleftrightarrow \mathrm{e}^{\alpha_i}_{(-m-1)}$$

preserves both the homogeneous grading and the color grading.

Unless stated otherwise, the variable q records the chosen grading (conformal in the conformal case and the auxiliary grading described above in the non-conformal case), while the variables x_i record the multigrading (or charge) associated with the generators.

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[1] It is possible to slightly relax this condition and consider graphs with loops [35].

We now return to the connection with arc algebras. The corresponding Hilbert series of $J_\infty(R_\Gamma)$ is what we call the *graph series*

$$H_{\Gamma,\mathbf{b}}(q) := \sum_{\mathbf{n}\in\mathbb{N}_0^r} \frac{q^{\frac{1}{2}\mathbf{n}A_\Gamma\mathbf{n}^{\mathsf{T}}+b_1n_1+\cdots+b_rn_r}}{(q)_{n_1}\cdots(q)_{n_r}},$$

where A_Γ is the (symmetric) adjacency matrix of a graph Γ with r vertices and $\mathbf{b}=(b_1,\dots,b_r)\in\mathbb{N}^r$. Here we use the standard q-Pochhammer notation

$$(q)_n=\prod_{j=1}^{n}(1-q^j) \qquad \text{and} \qquad (q)_\infty=\prod_{j=1}^{\infty}(1-q^j),$$

whenever the latter product is viewed formally or converges analytically. The vector $\mathbf{b}$ determines the grading of the generators of W_Γ. Equivalently, we can start from the *full* character of W_Γ,

$$H_\Gamma(q,\mathbf{x}) := \sum_{\mathbf{n}\in\mathbb{N}_0^r} \frac{\mathbf{x}^{\mathbf{n}}\, q^{\frac{1}{2}\mathbf{n}A_\Gamma\mathbf{n}^{\mathsf{T}}}}{(q)_{n_1}\cdots(q)_{n_r}},$$

and specialize to $\mathbf{x}=q^{\mathbf{b}}$. Essentially the same type of (q,x)-series appeared in so-called "knots-quivers" correspondences; see [23, 33].

In this paper, we are only interested in the standard grading $\mathbf{b}=(1,\dots,1)$, in which case

$$\deg\big(\mathrm{e}^{\alpha_i}_{(-m-1)}\big)=m+1,$$

and we simply write $H_\Gamma(q)$ instead of $H_{\Gamma,\mathbf{b}}(q)$, thus recovering the original graph series introduced in [15, 30, 31]. Although not directly related to this work, it is worth noting that there is an interesting connection among the Donaldson–Thomas invariants of symmetric quivers, cohomological Hall algebras, and the dual of the principal subspace of W_Γ [20, 21] (see also [22]).

Since we have encountered a q-zeta–type value $\mathfrak{z}_q(k)$, it is natural to ask whether other q-multiple zeta values also arise as graph series, possibly with additional Euler factors. As indicated above, we are primarily interested in the standard grading, corresponding to $b_i=1$. Within this framework, the q-MZVs relevant to our work were introduced by Ohno, Okuda, and Zudilin in [43]:

$$\mathfrak{z}^*_q(a_1,\dots,a_k) := \sum_{n_1\geq n_2\geq\cdots\geq n_k\geq 1} \frac{q^{n_1}}{(1-q^{n_1})^{a_1}\cdots(1-q^{n_k})^{a_k}}. \tag{0.1}$$

The star symbol indicates that the summation is over non-strict summation variables as opposed to the more standard model:

$$\mathfrak{z}_q(a_1,\dots,a_k) := \sum_{n_1>n_2>\cdots>n_k\geq 1} \frac{q^{n_1}}{(1-q^{n_1})^{a_1}\cdots(1-q^{n_k})^{a_k}}.$$

The difference between the strict and non-strict q-MZV models is controlled by the "lower depth" q-MZVs, so one can easily transform between the two. Our next result concerns graph series and multiple q-zeta values (0.1).

Theorem 0.1. *For every choice of positive integers $a_1,\dots,a_k$, there is a simple graph $Z_{a_1,\dots,a_k}$ such that*

$$H_{Z_{a_1,\dots,a_k}}(q)=\frac{q^{-1}\mathfrak{z}^*_q(a_1,\dots,a_k)}{(q)_\infty^{k+a_1+\cdots+a_k}}.$$

In particular, for $k=1$ and $a_1=r$, this recovers $H_{T_{2r+2}}(q)$ and Theorem 1.1 (ii).

0.1. **Graph series and multiple *q*-zeta series.** Now we are ready to explain how to construct all multiple q-zeta values in the framework of graph series and thus of vertex algebras. As already discussed in the introduction, we are interested in the q-MZV model given by

$$\mathfrak{z}_q^*(a_1,\dots,a_k)=\sum_{n_1\geq n_2\geq\cdots\geq n_k\geq 1}\frac{q^{n_1}}{(1-q^{n_1})^{a_1}\cdots(1-q^{n_k})^{a_k}}.$$

Consider a disconnected graph with $2k$ vertices $\{1,\dots,k,1',\dots,k'\}$ and k edges:

$$1\text{ - }1'\quad 2\text{ - }2'\quad 3\text{ - }3'\quad 4\text{ - }4'\quad 5\text{ - }5'\quad\cdots\qquad\cdots\qquad\cdots\qquad\cdots\qquad k\text{ - }k'$$

We next construct a graph, denoted by $Z_{a_1,\dots,a_k}$, which will contain the above graph as a subgraph. In this new graph, the nodes denoted by $1',\dots,k'$ are leaves (so they are only adjacent to $1,2,\dots,k$, respectively).

To construct our graph, we attach various simple star graphs to the underlying graph with $2k$ vertices. For better transparency, in parallel, we consider the graph $Z_{1,1,1,2,2}$ displayed on Figure 1.

We first attach a_1 simple $(k+1)$-star graphs to the k nodes enumerated by $1,2,\dots,k$ (e.g., on Figure 1, we attach one 6-star graph with the central node 6 to the nodes 1, 2, 3, 4, 5). Then we attach a_2 simple star graphs with k legs to the $k-1$ nodes enumerated by $1,2,\dots,k-1$ (e.g., on Figure 1, we attach one 5-star graph with central node 8 to the nodes 1, 2, 3, 4). We continue until we attach a_k 2-star graphs (i.e., paths of length two) to node 1 (e.g., on Figure 1, we attach two 2-star graphs with central vertices 16 and 18 to node 1).

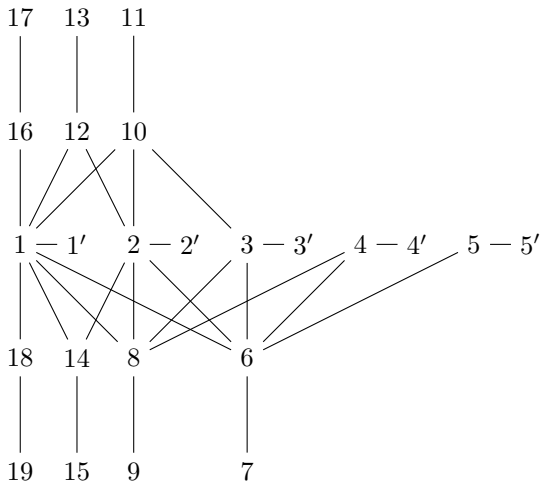


FIGURE 1. Graph $Z_{1,1,1,2,2}$.

The Hilbert series of $R_{Z_{a_1,\dots,a_k}}$ can be recursively computed using the relations

$$H_t(R_{Z_{a_1,\dots,a_k}})=H_t(R_{Z_{a_1-1,\dots,a_k}})+\frac{t}{(1-t)^{k+1}}\left(\frac{1+t}{1-t}\right)^{a_1+\cdots+a_k-1},$$
$$H_t(R_{Z_{0,\dots,0,a_i,a_{i+1},\dots,a_k}})=H_t(R_{Z_{a_i,\dots,a_k}}).$$

A closed formula for $H_t(R_{Z_{a_1,\dots,a_k}})$ is somewhat messy so we do not display it here.

For the graph series, we have the following result:

$$H_{Z_{a_1,\dots,a_k}}(q)=\frac{q^{-1}\mathfrak{z}_q^*(a_1,\dots,a_k)}{(q)_\infty^{a_1+\cdots+a_k+k}}.$$

Proof of Theorem 0.1. Again, the proof is easier to follow by considering an example in parallel; see Figure 1. We first sum over all $(k+1)$-star subgraphs. By definition, there are a_1 such subgraphs. For each subgraph, we start from the leaf node s (for instance, node 7 in Figure 1) and the central node t (for instance, node 6 in Figure 1). This leads to the following summation:

$$\sum_{n_s,n_t\geq 0}\frac{q^{n_s(n_t+1)+n_t(n_1+\cdots+n_k+1)}}{(q)_{n_s}(q)_{n_t}}=\sum_{n_t\geq 0}\frac{q^{n_t(n_1+\cdots+n_k+1)}}{(q^{n_t+1})_\infty(q)_{n_t}}=\frac{1}{(q)_\infty\big(1-q^{n_1+\cdots+n_k+1}\big)}.$$

Repeating this a_1 times (observe that the summation variables corresponding to the relevant subgraphs are independent) results with

$$\frac{1}{(q)_\infty^{a_1}\big(1-q^{n_1+\cdots+n_k+1}\big)^{a_1}}.$$

We repeat the same procedure for k-star graphs attached to $1,2,\dots,k-1$. These subgraphs contribute with

$$\frac{1}{(q)_\infty^{a_2}\big(1-q^{n_1+\cdots+n_{k-1}+1}\big)^{a_2}}.$$

We continue until we sum over paths of length two (e.g., nodes 17 and 16) attached to the node 1. Here the contribution is

$$\frac{1}{(q)_\infty^{a_k}\big(1-q^{n_1+1}\big)^{a_k}}.$$

We also have to sum over the variables that correspond to the "prime" nodes:

$$\sum_{n_1',\dots,n_k'\geq 0}\frac{q^{n_1'(n_1+1)+\cdots+n_k'(n_k+1)}}{(q)_{n_1'}\cdots(q)_{n_k'}}=\frac{1}{\big(q^{n_1+1}\big)_\infty\cdots\big(q^{n_k+1}\big)_\infty}.$$

Combining everything with the final k-fold summation over $n_1,\dots,n_k$, we quickly get

$$\frac{1}{(q)_\infty^{\sum_{i=1}^k a_i+k}}\sum_{n_1,\dots,n_k\geq 0}\frac{q^{n_1+\cdots+n_k}}{\big(1-q^{\sum_{i=1}^k n_i+1}\big)^{a_1}\big(1-q^{\sum_{i=1}^{k-1} n_i+1}\big)^{a_2}\cdots\big(1-q^{n_1+1}\big)^{a_k}}.$$

Setting $n_1+n_2+\cdots+n_k+1\to m_1$, $n_1+n_2+\cdots+n_k+1\to m_2,\dots,n_1+1\to m_k$ proves the claim. □

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